

Worksheet: Unsteady-state material balances for a two-phase, multicomponent system

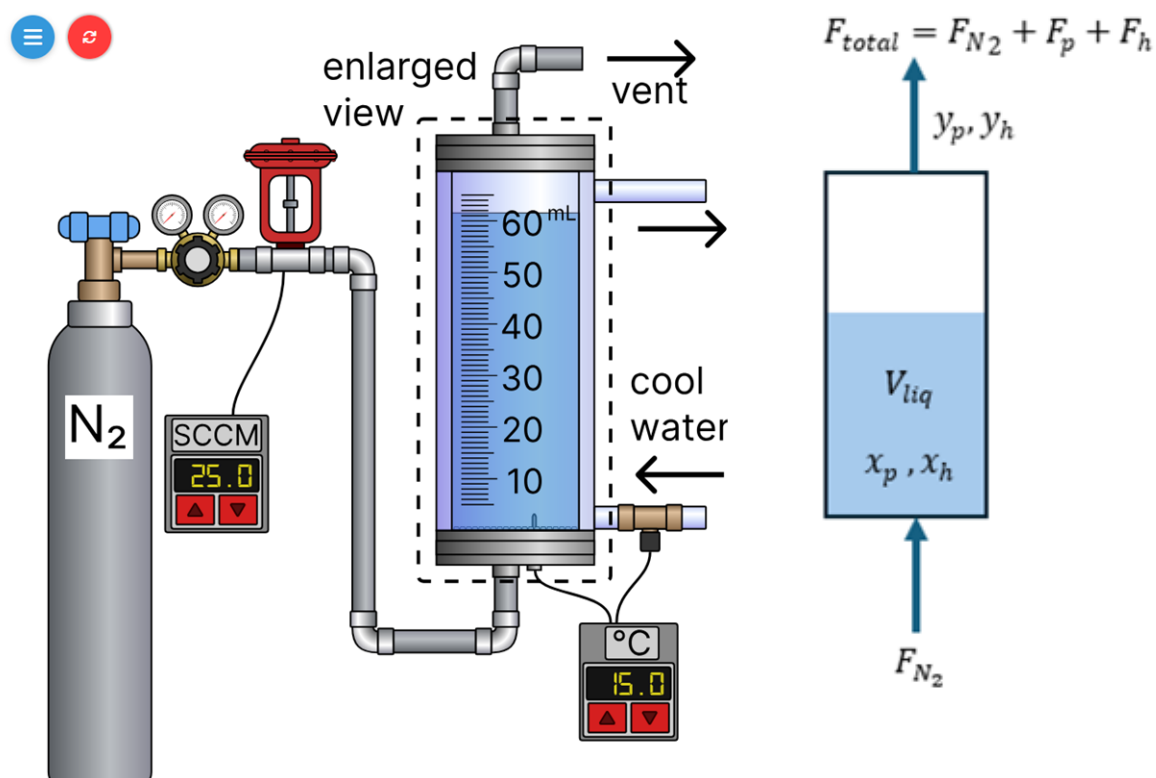
Name(s): _____

In this experiment, N_2 flows through a “bubbler” (a vapor-liquid contactor) that initially contains an equimolar liquid mixture of n-pentane and n-hexane. As N_2 passes through the bubbler, both components evaporate and leave with the exiting N_2 . Unsteady-state material balances and Raoult’s law are used to compare the measured and calculated changes of the liquid volume with time.

Student Learning Objectives

1. Define a control volume and write unsteady-state component mole balances for a semibatch system.
2. Relate the molar flow rate of an evaporating species to the rate of change of liquid level.
3. Apply Raoult’s law to calculate the composition and total flow rate of a multicomponent vapor leaving a bubbler.
4. Explain how temperature, carrier-gas flow rate, and changing liquid composition affect evaporation.

Equipment and flow representation



The bottom of the vessel contains a diffuser, so bubbles are evenly distributed across the vessel diameter. The entire system operates near atmospheric pressure, although there is some pressure drop between the inlet and outlet.

Unsteady-state material balances

Notation

x_i = mole fraction of component i in the liquid phase

y_i = mole fraction of component i in the vapor

P_{gauge} = gauge pressure (bar)

P_{inlet} = inlet pressure (bar)

P_{outlet} = outlet pressure (1.00 bar)

P_i = partial pressure of component i (bar)

P_i^{sat} = saturation pressure of component i (bar)

v = N₂ volumetric flow rate (SCCM, standard cm³/min)

F_{N_2} = N₂ molar flow rate (mol/min)

F_i = molar flow rate of component i in effluent (mol/min)

F_{total} = total molar flow rate (mol/min)

MW_i = molecular weight of component i (g/mol)

ρ_i = liquid density of component i (g/mL)

V_{liq} = volume of liquid remaining in bubbler (mL)

V_p = volume of liquid that is n-pentane (mL)

V_h = volume of liquid that is n-hexane (mL)

$\dot{V}_p$ = rate of volume change of n-pentane liquid (mL/min)

$\dot{V}_h$ = rate of volume change of n-hexane liquid (mL/min)

$\dot{V}_{liq}$ = rate of volume change of liquid (mL/min)

N_i = moles of component i in the bubbler (mol)

t = time (s)

Physical properties

Property	n-pentane	n-hexane
MW (g/mol)	72.2	86.2
ρ_i (g/cm ³)	0.63	0.66
Antoine A	3.9892	4.0027
Antoine B	1070.6	1171.5
Antoine C	-40.454	-48.784

Assumptions

- The vapor leaving the bubbler is in equilibrium with the liquid in the bubbler.
- n-Pentane and n-hexane are chemically similar and form an ideal solution.
- Nitrogen does not dissolve appreciably in the liquid.
- The bubbler is isothermal.
- Nothing condenses downstream; all tubing and the vent are at 25 °C or above.
- The pressure in the bubbler is essentially $P_{\text{outlet}} = 1.00$ bar.
- Liquid volumes are additive (no excess volume of mixing).
- Liquid densities do not change significantly over the temperature range from 15 to 25°C.
- Standard conditions for gas flow (SCCM) are at 25 °C and 1.0 bar.

Before running the experiment:

1. The bubbler can be set to one of three temperatures: 15, 20, or 25°C. All tubing downstream of the bubbler (and the vent) are at a temperature of at least 25°C. The change in liquid level in the bubbler is measured over time. As discussed below, this can be related to the molar flow rates of n-pentane and n-hexane out of the bubbler.
2. Bubblers are often assumed to operate close to vapor-liquid equilibrium. Furthermore, because n-pentane and n-hexane are chemically similar, we assume they form an ideal solution. Thus, Raoult's Law determines the gas-phase pressure of each compound. Write the equations to determine the partial pressures of n-pentane and n-hexane exiting an equilibrated bubbler.
3. How will operating the bubbler at higher temperatures affect the flow rate coming out of the bubbler? How will it affect the vapor composition?
4. As the experiment proceeds, the composition of liquid remaining in the bubbler changes. How do you expect it to change? How will the change in composition affect the flow rate of the vapor exiting the bubbler over time?

Unsteady-state material balances

5. Is a plot of volume in bubbler versus time linear? If slope changes with time, explain why.

Running the experiment:

1. The tank initially contains a 50/50 molar mixture of n-pentane and n-hexane.
2. The bubbler temperature set point is initially at 15°C.
3. Select a N₂ flow rate setting ν between 20.0 and 40.0 SCCM (standard cm³/min) and record the value at the top of Table 1.
4. Click on the N₂ cylinder valve to open it.
5. Adjust the cylinder regulator to a gauge pressure P_{gauge} greater than zero to start N₂ flow through the bubbler. Record the gauge pressure at the top of Table 1 and record the absolute pressure P at the top of Table 1.
6. Record in Table 1 the liquid volume in the bubbler initially and at different time points (every time the liquid volume changes by 3 or 4 mL), until the liquid level has declined to ~20% of its initial value. The system automatically shuts down when that level is reached. Below that level, the assumption of vapor-liquid equilibrium is not likely to hold due to the small volume of liquid in the container.
7. To complete this experiment in a reasonable time, the simulation accelerates time by a factor of 60. Thus, when recording time in the worksheet, measurements in seconds in the simulation correspond to minutes in the physical experiment.
8. Reset the system to refill the bubbler.
9. Repeat this process, starting with step 2, for temperatures 20°C and 25°C and record measurements in Table 1. Use the same N₂ flow rate and the same cylinder gauge pressure.

Unsteady-state material balances

Table 1 Volume and time measurements

$v = \text{_____}$ $P_{gauge} = \text{_____}$ $P = \text{_____}$

15°C		20°C		25°C	
t (min)	V_{liq} (mL)	t (min)	V_{liq} (mL)	t (min)	V_{liq} (mL)

Plot liquid volume versus physical time for the three experiments. What patterns do you see?

Unsteady-state material balances

Analysis:

1. Use material balances and Raoult's law to compute the expected rate of volume decrease. This requires several steps. The equations for the composition of the vapor leaving the bubbler, assuming vapor-liquid equilibrium, are:

$$y_i P_{outlet} = P_i = x_i P_i^{sat}$$

where P_i is the partial pressure of n-pentane or n-hexane,
 y_i is the mole fraction in the vapor.

P_{outlet} is the total outlet pressure (1.00 bar)

x_i is the liquid mole fraction

P_i^{sat} is the saturation pressure at a given temperature.

We can write one equation for n-pentane and one for n-hexane. Values for P_i^{sat} are calculated by the Antoine equation, and P_i^{sat} values are fixed once the bubbler temperature is set.

2. Next, determine the molar flow rate of each component exiting the bubbler. Calculate the N_2 molar flow rate (F_{N_2}) using the ideal gas law:

$$F_{N_2} = \frac{1.01v}{R(273+25)}$$

where the standard conditions are 1.01 bar and 25°C. Because we do not expect N_2 to accumulate in the vessel, this is also the molar flow rate of N_2 exiting the bubbler.

3. The total pressure exiting the bubbler $P_{outlet} = 1.0$ bar. The partial pressure of N_2 is equal to this pressure minus the partial pressures of n-pentane (P_p) and n-hexane (P_h):

$$P_{N_2} = P_{outlet} - P_p - P_h = P_{outlet} - x_p P_p^{sat} - x_h P_h^{sat}$$

Thus, the molar flow rate of n-pentane exiting the bubbler is equal to the flow rate of N_2 through the bubbler, multiplied by the ratio of the n-pentane to N_2 vapor phase mole fractions, i.e. :

$$y_p = \frac{F_p}{F_{tot}}$$

$$y_{N_2} = \frac{F_{N_2}}{F_{tot}}$$

Dividing these two equations and rearranging:

$$F_p = F_{N_2} \frac{y_p}{y_{N_2}} = F_{N_2} \frac{P_p}{P_{N_2}}$$

Then, the molar flow rate of each component is:

Unsteady-state material balances

$$F_p = F_{N_2} \frac{x_p P_p^{sat}}{P_{outlet} - x_p P_p^{sat} - x_h P_h^{sat}}$$
$$F_h = F_{N_2} \frac{x_h P_h^{sat}}{P_{outlet} - x_p P_p^{sat} - x_h P_h^{sat}}$$

From these equations, we can determine the rate of mass flow out of each component from the molecular weight. And we can relate this to the drop in liquid level by dividing by the liquid density:

A material balance on each component is: Accumulation = flow in – flow out

The flow rates in of n-pentane and n-hexane are zero so the material balances become

$$\frac{dV_p}{dt} = \dot{V}_p = F_p \quad \frac{dV_h}{dt} = \dot{V}_h F_h$$
$$\dot{V}_p = \frac{(MW)_p}{\rho_p} F_{N_2} \frac{x_p P_p^{sat}}{P_{outlet} - x_p P_p^{sat} - x_h P_h^{sat}}$$
$$\dot{V}_h = \frac{(MW)_h}{\rho_h} F_{N_2} \frac{x_h P_h^{sat}}{P_{outlet} - x_p P_p^{sat} - x_h P_h^{sat}}$$

The total rate of volume change $\dot{V}_{liq}$ is equal to the sum of these two rates of change of liquid volume ($\dot{V}_p, \dot{V}_h$), which can be compared to the experimental data.

4. By plugging in values of $x_p = x_h = 0.50$, we can determine the initial rate of volume decline. Try plugging in appropriate numbers for the other parameters. Does your answer agree with the slope at early times on your plot?

In the following section, the complete time-dependent solution is obtained by numerically solving the differential mass balances using Python or other software.

5. The composition of the remaining liquid changes with time because n-pentane is removed from the system faster than n-hexane. (Why is that?) To account for this, we need to write an unsteady-state mole balance:

$$\frac{dN_p}{dt} = F_{N_2} \frac{x_p P_p^{sat}}{P_{outlet} - x_p P_p^{sat} - x_h P_h^{sat}}$$

where $x_p = \frac{N_p}{N_p + N_h}$

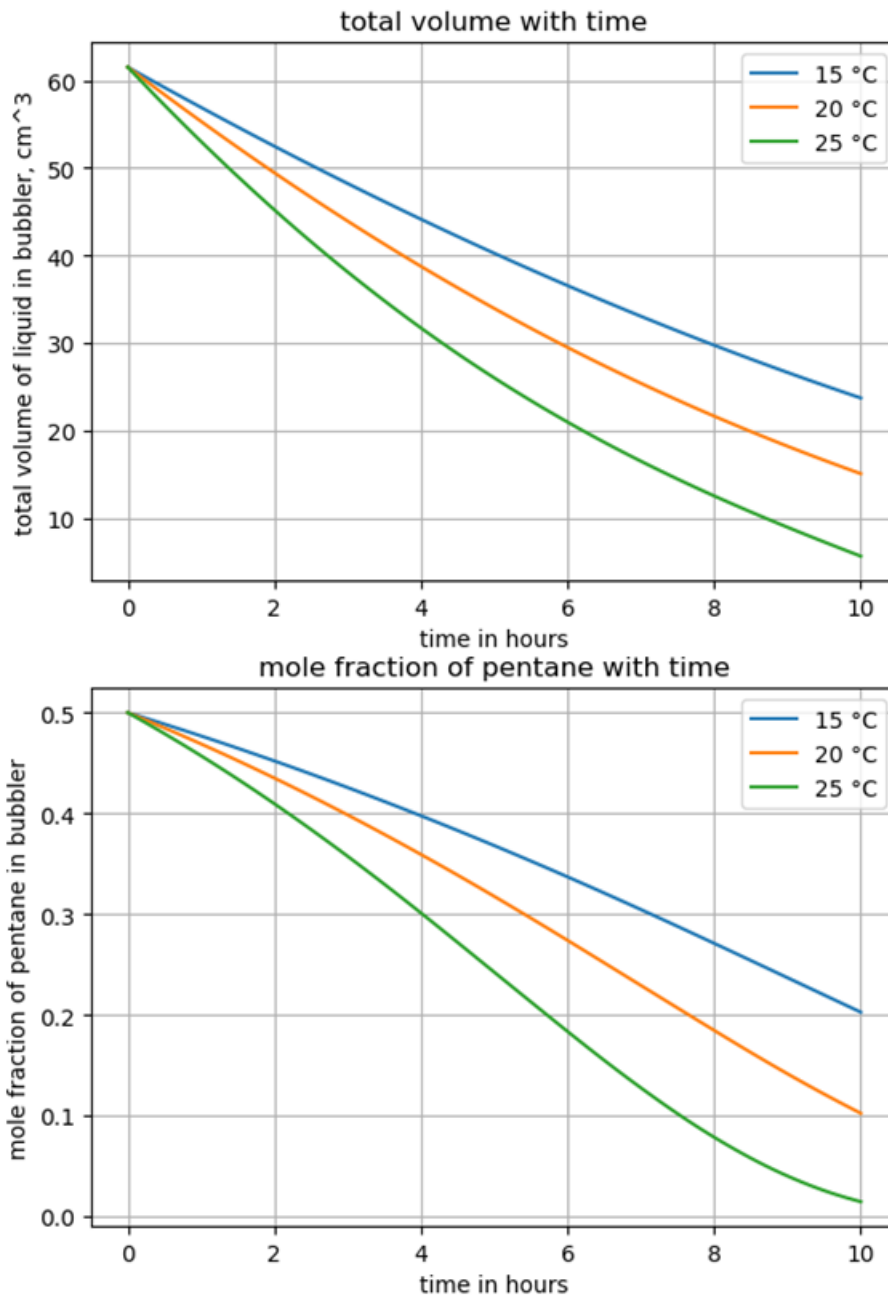
A similar equation for n-hexane is:

Unsteady-state material balances

$$\frac{dN_h}{dt} = F_{N_2} \frac{x_h P_h^{sat}}{P_{outlet} - x_p P_p^{sat} - x_h P_h^{sat}}$$

These two differential equations must be solved simultaneously given a known initial value, i.e., this is an *initial value problem*. Solving these differential equations numerically gives the number of moles of each component as a function of time. Using the molecular weight and density of the two components, this can be converted into total liquid volume as a function of time for comparison to the experimental data.

Python plot: Data from the digital experiment could be plotted to compare to the Python results below. Note that these plots are for $v = 35$ SCCM, so your results will differ somewhat.



Unsteady-state material balances

Questions to answer

1. Why does $\frac{dV_{liq}}{dt}$ change with time?
2. Does the nitrogen mole fraction in the outlet remain constant? Explain.
3. Does the more volatile component always leave at a greater molar rate? Explain/
4. Why does total outlet flow rate decrease as the liquid becomes richer in n-hexane?